

Python: an introduction

- program structure: constants, acquisition, computation, visualisation
- variables and constants
- `input()` and `print()`
- `import` statement to access existing python libraries, for instance `import math`: to be put at the beginning of our program

Details

- `input()` always returns a string, a sequence of character, therefore if that data represents a value (integer or real), it needs to be converted, by means of `int()` or `float()`.
- given a value (integer or real) it is possible to covert and obtain a string consisting of the numerical characters (and the "." separator) representing the value into a string, by means of the `str()` function.

Exercises in class

Exercise 1

Write a Python program that receives in input a positive integer value (it is surely so) representing the side of a square and computes the area. Finally, it shows the result.

Exercise 2

Write a Python program that receives in input a positive integer value (it is surely so) representing the radius of a circle and it computes the length of the circumference and its area. Finally, it shows the results.

Initial version with constant values (e.g., 3.14159) distributed in the code.

Note: avoid putting constants directly in the code. It reduces readability and flexibility.

Variant with constant at the top (written in capital letters).

Note: for established values it is possible to *think* that there is a library available. This is the case of the `math` library. Search python documentation to find that a `math.pi` exists.

Exercise 3

Write a python program that asks for a positive integer, representing a time expressed in seconds; the program converts it into days, hours, minutes, and seconds, and finally shows the result. *To be completed in class.*

Proposed exercises

Proposed exercise 1

Write a python program that, after receiving an integer value, calculates and displays 1 if the value is odd, or 0 otherwise. Make a variation that does not use the remainder (modulus) operator.

Proposed exercise 2

Create a program to split a restaurant bill. The program receives in input the total bill amount and the number of people sharing it. Apply a fixed tip percentage constant, then output the total tip, the total amount due, and each person's share.

Proposed exercise 3

Write a program that receives in input a real value representing a temperature expressed in Celsius and computes and visualizes the same temperature expressed in Fahrenheit. The formula for the relationship is $c = (f - 32) * 5/9$.

Proposed exercise 4

Write a program that receives in input a real value representing a temperature expressed in Fahrenheit and computes and visualizes the same temperature expressed in Celsius.

Proposed exercise 5

Write a program that receives an integer value in input and a real one corresponding to height (in cm) and weight (in kg), respectively. The program computes and shows the BMI of the person. The BMI is calculated by dividing the weight in kilograms by their height in metres squared.

Proposed exercise 6

Write a program that receives in input the height of a person expressed in centimeters and computes the ideal weight according to the BMI formula (use $BMI = 22$)